

TEACHER MASTER KEY: GENERAL & CONCEPTUAL PHYSICS

Kinematics Graphing Challenges 1 & 2 (Periods 1–6)

TEACHER KEY • MR. MUDRY

Challenge 1: Harbor Boat & Supply Drone Rendezvous

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Plotted Points on Grid:

- **Boat:** (0s, 300m), (10s, 360m), (20s, 420m), (30s, 480m), (40s, 540m), (50s, 600m), (55s, 630m), (60s, 660m).
- **Drone:** (0 to 20s flat on dock at 0m), (30s, 180m), (40s, 360m), (50s, 540m), (55s, 630m), (60s, 720m).

1. **Rendezvous Coordinates:** Time $t = 55$ seconds • Position $x = 630$ meters from dock.
2. **Steepness & Speed:** The **Drone line is steeper**. A steeper slope means a faster speed (covers more meters each second).
3. **Actual Drone Flight Duration:** $55\text{ s} - 20\text{ s delay} = 35$ seconds in the air.
4. **Battery Range Check:** Round-trip distance = $630\text{ m out} + 630\text{ m back} = 1,260$ meters.
→ Since max battery range is $1,200\text{ m}$, **No, the drone cannot make it back** (crashes 60 m short of the dock).

Challenge 2: Highway Courier Journey & Patrol Intercept

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Plotted Points on Grid:

- **Delivery Truck:** (0s, 0m) → (20s, 300m) → (50s, 300m, flat line) → (90s, 1,300m).
- **Patrol Line A:** Starts at (30s, 0m), connects to rest stop at (50s, 300m).
- **Patrol Line B:** Starts at (30s, 0m), connects to highway end at (90s, 1,300m).

1. **Understanding Rest Stop:** The position stays at 300 m while time continues to tick forward (slope = 0 m/s). This proves the truck is stationary (stopped).
2. **Patrol Car Speed Calculations:**
 - **Option A (Rest Stop at $t = 50\text{ s}$):** Speed = $300\text{ m} / 20\text{ s} = 15\text{ m/s}$.
 - **Option B (Highway End at $t = 90\text{ s}$):** Speed = $1,300\text{ m} / 60\text{ s} = 21.7\text{ m/s}$.
3. **Patrol Safe Speed Limit (Max Limit = 20 m/s):**
 - Option A needs 15 m/s ($15 \leq 20\text{ m/s}$) → **SAFE & POSSIBLE** ✓
 - Option B needs 21.7 m/s ($21.7 > 20\text{ m/s}$) → **EXCEEDS SPEED LIMIT** ✗→ The patrol car must choose **Option A** to intercept at the rest stop.